



Enhancing strength and ductility of Nb alloy via Y addition by vacuum electromagnetic levitation melting under oxygen carrier route

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ABSTRACT

Niobium (Nb) is considered as candidate material for heat pipe in space nuclear reactors due to high melting point, low density, excellent compatibility with alkali liquid metals, irradiation resistance, workability and welding performance, but low strength restricts its engineering application. In this work, yttrium (Y) was introduced into Nb matrix to achieve solution strengthening from Y atoms and dispersion strengthening from yttrium oxides. The latter one was realized by oxygen carrier route and vacuum electromagnetic levitation melting technique. Finally, one Nb521 (Nb-5.0 wt%W-1.7 wt%Mo-1.0 wt%Zr-0.07 wt%C) and four kinds of Nb521-xY (x = 0.1, 0.2 0.3 and 0.4 wt%) alloys were prepared. TEM characterization indicated that 80 nm Y₂O₃ particles were formed in Nb521-0.1Y, which exhibited the highest yield strength of 673 MPa, ultimate tensile strength of 740 MPa as well as larger elongation of 12.8%. Moreover, the solution strengthening of Y atoms and Orowan strengthening of yttrium oxides were evaluated quantitatively.

1. Introduction

Niobium (Nb) is considered to be one of the most promising candidates for heat pipe in space nuclear reactors due to its excellent properties, such as high melting point (2468 °C), low density (8.57 g/cm³), excellent compatibility with alkali liquid metals, irradiation resistance of fast neutron, workability and welding performance [1–3]. However, some disadvantages like low recrystallization temperature, tensile strength and creep resistance, seriously restrict the engineering application of Nb alloy in the heat pipe of space reactor [4–6]. For example [3], the recommended temperature window of Nb-1wt.%Zr was about 800–1250 K. The lower temperature limit of 800 K was determined by irradiation brittleness of fast neutron, and the temperature upper limit of 1250 K was influenced by recrystallization brittleness. Furthermore, the recommended stresses at 800–1100 K and 1100–1250 K were only about 68–74 MPa and 20–74 MPa, which were determined by 1/3 ultimate tensile strength and creep strength, respectively. Thus, it's necessary to improve the strength of Nb alloy. Usually, the strength of Nb alloy can be

increased by dispersion strengthening and solution strengthening. In PWC-11 (Nb-1wt.%Zr-0.1 wt%C), the dispersion strengthening was mainly from Nb–Zr–C carbides [7]. In C-103 (Nb-10 wt%Hf-1wt.% Ti-0.7 wt%Zr), the solution strengthening was mainly from Zr and Hf as well as trace W and Mo [8].

Atomic radius of yttrium (Y) is 1.81 Å, which is larger than that of Nb (1.43 Å). Thus, introducing Y into Nb matrix can improve the strength of Nb alloy by lattice distortion theoretically. Furthermore, yttrium oxides possess significantly low Gibbs free energy and can be acted as dispersion strengthening phases [9]. Recently, oxide dispersion strengthening (ODS) was achieved in other refractory metals and steels. In Ref. [10], ODS-Mo exhibited yield strength (YS) over 800 MPa and tensile elongation as large as ~ 40% at room temperature by introducing nanometric oxide particles uniformly distributed in the grain interior. In Ref. [11], ODS-W presented YS of 1210 MPa at room temperature and the nil ductility temperature is below 25 °C. In Ref. [12], ODS-Steel displayed significant low creep rates and high strength (>2 GPa) as well as large tensile ductility (RA > 40%) by introducing high number

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density Y–Ti–O nanoprecipitates into matrix. Therefore, it is possible to increase the strength and ductility of Nb alloy simultaneously by ODS route, but the size of yttrium oxides should be in nano scale.

All the above mentioned ODS-Mo, ODS-W and ODS-Steel were prepared by powder metallurgy method. But Nb alloy are usually produced via vacuum melting technique at present. It is difficult to manufacture ODS alloy in liquid metal because the nano yttrium oxides usually possess high interface energy and poor wettability with molten metal, thereby suffer coarsening and agglomeration during melting [13]. To solve this problem, the oxygen carrier route was developed and the mechanism was the in-situ reaction between elemental yttrium and activated oxygen from carrier during rapid solidification. In Refs. [14–16], TiO_2 and Fe_2O_3 were selected as oxygen carriers and FeNiMn-ODS, 9Cr-ODS steels were prepared successfully.

In this work, the oxygen carrier route was adopted and achieved by vacuum electromagnetic levitation melting technique. On the one hand, ZrO_2 was selected as the oxygen carrier and its melting point (2700 °C) was slightly higher than that of Nb. On the other hand, the water-cooled copper crucible can accelerate the solidification process. Besides, except for Y and ZrO_2 , 5.0 wt% W, 1.7 wt% Mo, 1.0 wt% Zr and 0.07 wt% C were also added to achieve solution strengthening from W, Mo, Zr, Y atoms and second phase strengthening from carbides and oxides, simultaneously. Finally, one Nb521 (Nb-5.0 wt%W-1.7 wt%Mo-1.0 wt%Zr-0.07 wt%C) and four kinds of Nb521-xY (x = 0.1, 0.2, 0.3 and 0.4 wt%) alloys were prepared. The results indicated that Nb521-0.1Y exhibited the highest YS of 673 MPa, ultimate tensile strength (UTS) of 740 MPa as well as larger elongation of 12.8%. And the solution strengthening of Y and Orowan strengthening of yttrium oxides were evaluated quantitatively.

2. Material preparation and characterization

2.1. Materials and manufacturing procedure

Firstly, the master alloy ingot of Nb521 was produced via vacuum electron beam melting technique and supplied by Ningxia Orient Tantalum Industry Co., Ltd. Then the Nb521 master ingot was cut into cube pieces with the size of 15 mm. Secondly, 1.3 kg Nb521 master ingot pieces and some Y pieces (>99.99 wt%) as well as some ZrO_2 pieces (>99.99 wt%) were placed in the crucible. 1.3 g Y and 1.3 g ZrO_2 , 2.6 g Y and 2.6 g ZrO_2 , 3.9 g Y and 3.9 g ZrO_2 , 5.2 g Y and 5.2 g ZrO_2 were added to prepare Nb521-0.1Y, Nb521-0.2Y, Nb521-0.3Y, Nb521-0.4Y ingots, respectively. Finally, vacuum electromagnetic levitation melting was adopted to prepare the Nb521 and Nb521–Y ingots and the main parameters included DC current of 90 A, DC voltage of 550 V, vacuum degree of 5×10^{-3} Pa, AC voltage of 250 V and induced current of 200 A. In this process, yttrium powder provides the source of [Y], and ZrO_2 provides the source of [O]. Compared to other yttrium-containing compounds, Y_2O_3 has the minimum Gibbs free energy. Therefore, [Y] will immediately react with [O] to form Y_2O_3 during levitation melting [15].

The compositions were determined by inductively coupled plasma-optic emission spectroscopy (ICP-OES) and infrared absorption analyses. Table 1 shows the measured compositions of Nb521 ingot and Nb521–Y ingots with various Y additions. Contents of W and Mo in all

the ingots were close to the designed values. The lower contents of Zr, C and Y can be resulted from that large ceramic phase were generated and further formed slags.

Subsequently, all the ingots were subjected to warm rolling and the rolling temperature was 400 °C. The total thickness reduction of about 50% was obtained through 10 passes.

2.2. Characterization

Microstructure was characterized via a laser scanning confocal microscope (LSCM, OLYMPUS) and a field-emission scanning electron microscope (FE-SEM, GeminiSEM 500) equipped with energy dispersive spectrometer (EDS, Esprit Compact) on the samples etched by 50 vol% H_2SO_4 + 30 vol% HNO_3 + 20 vol% HF. Besides, the samples were ground to a thickness of 40 μm and then subjected to ion thinning for transmission electron microscopy (TEM, Talos F200s) characterization. The phase structure was identified by X-ray diffractometry (XRD, MiniFlex600).

Tensile test was conducted on the INSTRON-5966 machine at room temperature with a strain rate of 0.001 s^{-1} . The tensile specimens were cut along the rolling direction and the geometry is presented in Fig. 1. The thickness of the tensile specimen was 1 mm. The tensile tests were carried out 3 times for each sample to ensure the precision.

3. Results

3.1. Grain size and carbide second phase

In order to evaluate the effect of Y addition on grain size better, Nb521 and Nb521–Y ingots were subjected to LSCM characterization. Fig. 2(a–e) shows the LSCM images of Nb521, Nb521–Y and Fig. 2(f) presents the variation curve of grain size with Y addition. The grain size was 423, 152, 103, 187 and 224 μm for Nb521, Nb521-0.1Y, Nb521-0.2Y, Nb521-0.3Y and Nb521-0.4Y, respectively. Fig. 3(a–e) shows the LSCM images of Nb521, Nb521–Y after deformation, and Fig. 3(f) presents the variation curve of average grain width with Y addition. Obviously, Y addition facilitated the grain refinement of Nb521. Furthermore, Nb521-0.1Y and Nb521-0.2Y displayed smaller grain size. As the addition contents of Y were increased to 0.3 and 0.4 wt%, the grain size became larger, which can be related to slighter heterogeneous nucleation. It has been proven that more Y additions would lead to higher yield rate of Y [17], which may further result in the increase of size and decrease of number density for Y-bearing particles. This phenomenon will be presented in 3.2 section. Less and larger particles would reduce the effect of heterogeneous nucleation thereby led to the grain growth, i.e., larger grain size in Nb521-0.3Y and Nb521-0.4Y.

Considering that Nb521 and Nb521–Y ingots exhibited close contents of C and Zr, thus the carbides was only characterized in Nb521. Fig. 4(a) shows the high-magnification LSCM image of Nb521. A large number of needle-like second phases were observed and their average length was about 20 μm . Fig. 4(b) shows the bright-field TEM image of Nb521 and the width of the typical needle-like second phase was about 120 nm. The inserted image in Fig. 4(c) displays the corresponding fast

Table 1

Measured compositions of Nb521 and Nb521–Y ingots with various Y additions, wt.%.

Sample	W	Mo	Zr	C	Y	Nb
Nb521	5.07	1.74	0.67	0.034	–	Bal.
Nb521-0.1Y	4.97	1.73	0.68	0.034	0.07	Bal.
Nb521-0.2Y	5.12	1.75	0.66	0.030	0.13	Bal.
Nb521-0.3Y	5.08	1.74	0.66	0.039	0.18	Bal.
Nb521-0.4Y	5.24	1.78	0.66	0.037	0.37	Bal.

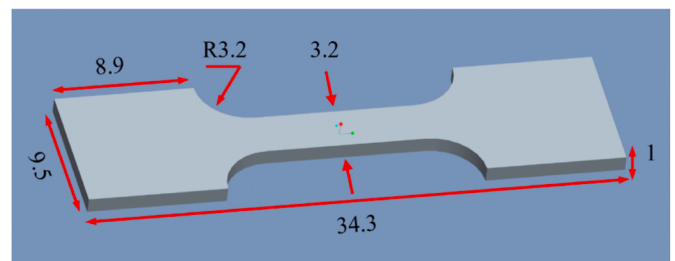


Fig. 1. Schematic drawing of the tensile specimen (unit in mm).

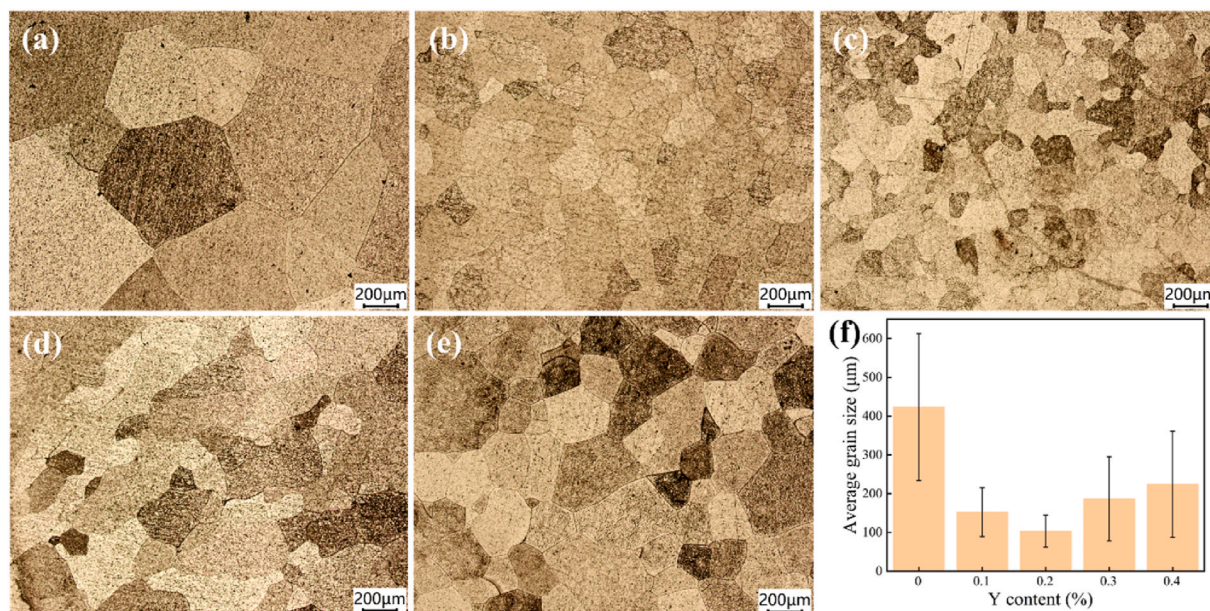


Fig. 2. LSCM images and grain size of Nb alloy: (a) Nb521, (b) Nb521-0.1Y, (c) Nb521-0.2Y, (d) Nb521-0.3Y, (e) Nb521-0.4Y, (f) variation curve of average grain size with Y addition.

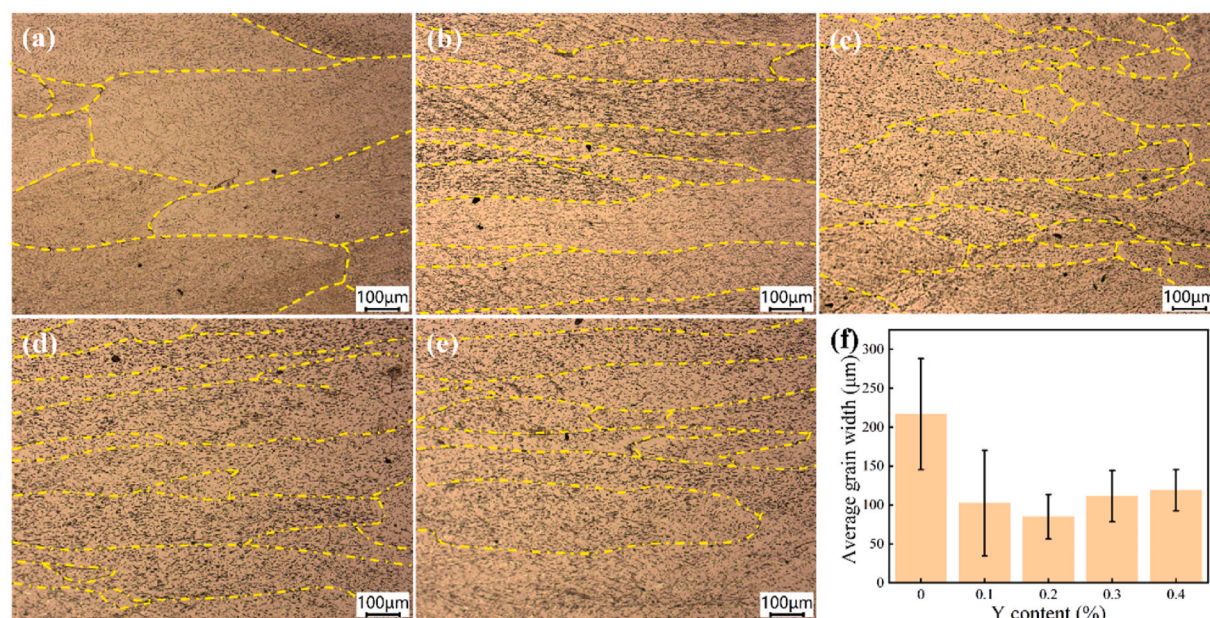


Fig. 3. LSCM images and grain size in the width direction of the Nb alloy rolled plate: (a) Nb521, (b) Nb521-0.1Y, (c) Nb521-0.2Y, (d) Nb521-0.3Y, (e) Nb521-0.4Y, (f) variation curve of average grain width with Y addition after deformation.

fourier transform (FFT) image of the second phase. The measured interplanar spacings of {200}, {220}, and {020} were 2.27, 1.62, and 2.27 Å, respectively, and the corresponding intersection angles of α_{12} , α_{13} , and α_{23} were about 46.7°, 93.4°, and 46.7°, which were close to the theoretical values of NbC as shown in Table 2. EDS mapping was performed in Fig. 4(d) to further characterize the second phase. The elemental distribution maps show that there is an obvious enrichment of the C element in the second phase, and no obvious W, Mo, Zr, or Y enrichment was found in it. In addition, due to the high content of Nb in the matrix, the low contrast of Nb is shown in Fig. 4(d). Thus, it can be inferred that the carbides in Nb521 and Nb521–Y were NbC phase.

3.2. Micron and nano scale Y_2O_3 second phase

Fig. 5 shows the SEM of Nb521–Y with various Y additions. For Nb521-0.1Y, some spherical second phases were observed and their size was about 1.6 μm as shown in Fig. 5(a). For the Nb521-0.2Y in Fig. 5(b), the size of these spherical second phases was increased to about 1.9 μm. For the Nb521-0.3Y in Fig. 5(c), the average size of spherical second phases further increased to about 2.5 μm. Moreover, the distance of some particles became closer and presented dumbbell shape and rod-like. When the content of Y addition was increased to 0.4 wt% as shown in Fig. 5(d), on the one hand, the average size of spherical second phases increased to about 2.9 μm. On the other hand, some large acicular second phases were formed along the grain boundaries.

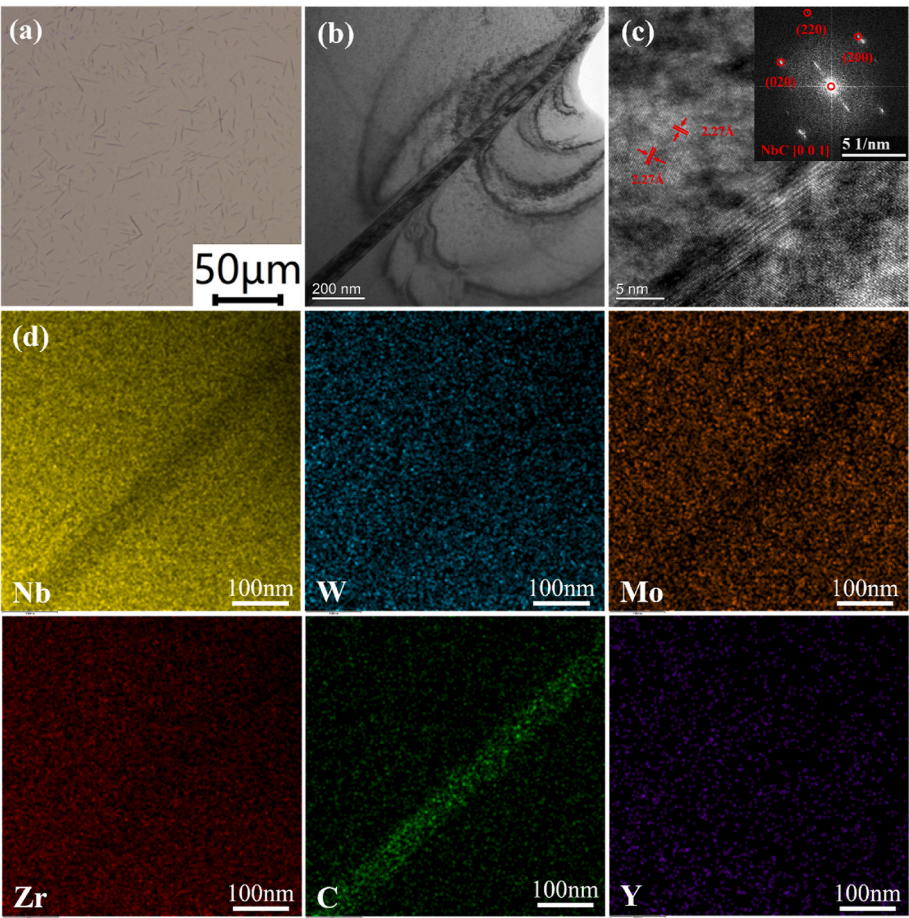


Fig. 4. (a) High-magnification LM image of Nb521, (b) BF-TEM image of Nb521 and (c) HR-TEM and the corresponding FFT image of second phase, (d) EDS mappings of Nb, W, Mo, Zr, C, and Y elements.

Table 2
Measured and theoretical interplanar spacings, intersection angle of NbC.

d(Å), α (°)	d ₁ {200}	d ₂ {220}	d ₃ {020}	α ₁₂	α ₁₃	α ₂₃
Measured	2.27	1.62	2.27	46.7	93.4	46.7
Theoretical	2.23	1.58	2.23	45	90	45

To confirm the compositions of the spherical and acicular second phases, EDS analysis was conducted on the phases marked in Fig. 5(a) and (d) and the results are presented in Fig. 6. EDS result of the spherical second phase in Fig. 6(a) is shown in Fig. 6(c), except for the matrix composition of Nb and C, peaks of Y and O elements were observed. EDS result of the acicular second phase in Fig. 6(b) is shown in Fig. 6(d), and similar composition characteristics was presented. Thus, it can be inferred that both the spherical and the acicular second phases were Y₂O₃.

Taking into account that the micron-scale Y₂O₃ became larger in Nb521-0.2Y, Nb521-0.3Y and even presented large acicular shape in Nb521-0.4Y, thus TEM characterization was only conducted on the Nb521-0.1Y to explore whether nano-scale yttrium oxides were formed. Fig. 7 shows the TEM results of Nb521-0.1Y sample. Some particles with the size of 80 nm were located along grain boundary under bright field image as shown in Fig. 7(a). The selected area electron diffraction (SAED) of marked particle in Fig. 7(a) is shown in Fig. 7(b). Except for the diffraction spots of niobium matrix, the diffraction spots of Y₂O₃ were also detected. The measured and theoretical interplanar spacings of {3 1̄ 2}, {611}, {321̄} and the corresponding intersection angles are presented in Table 3. Besides, TEM mapping images shown in Fig. 7(c)

also indicated that these particles were rich-in Y and O elements.

3.3. Tensile properties

Tensile properties of Nb521 and Nb521–Y at room temperature are shown in Table 4. The YS, UTS and elongation of Nb521 were 636 MPa, 680 MPa and 10.8%, respectively. Compared to Nb521, both Nb521-0.1Y and Nb521-0.2Y exhibited higher YS, UTS and larger elongation, simultaneously. Furthermore, Nb521-0.1Y showed the highest YS of 673 MPa, UTS of 740 MPa and excellent elongation of 12.8%. For the Nb521-0.3Y, although higher UTS was achieved, but lower YS, especially smaller elongation was also presented. When the content of Y addition reached 0.4 wt%, almost the lowest strength and the minimum elongation were found.

Nb521-0.1Y and Nb521-0.2Y displayed higher strength and larger elongation simultaneously can be attributed to the uniformly distributed yttrium oxides as shown in Fig. 5(a and b) and Fig. 6. Nb521-0.3Y and Nb521-0.4Y exhibited lower strength, especially smaller elongation can be resulted from the large spherical or acicular Y₂O₃ as shown in Fig. 5(c and d). As shown in Fig. 5, the oxide particles in Nb alloys with higher Y content are larger in size and tend to be distributed at grain boundaries, which leads to poor deformation coordination [17]. The stress at the particle/matrix interface is higher than that at the grain boundary and sub-grain boundary, resulting in a greater stress concentration at the interface. During the tensile test process, if the stress exceeds the ultimate tensile strength, it will lead to interface debonding and crack initiation [18]. The crack source then continues to expand along the particle/matrix interface, resulting in the mechanical properties rapidly deteriorating with the addition of Y greater than 0.1%.

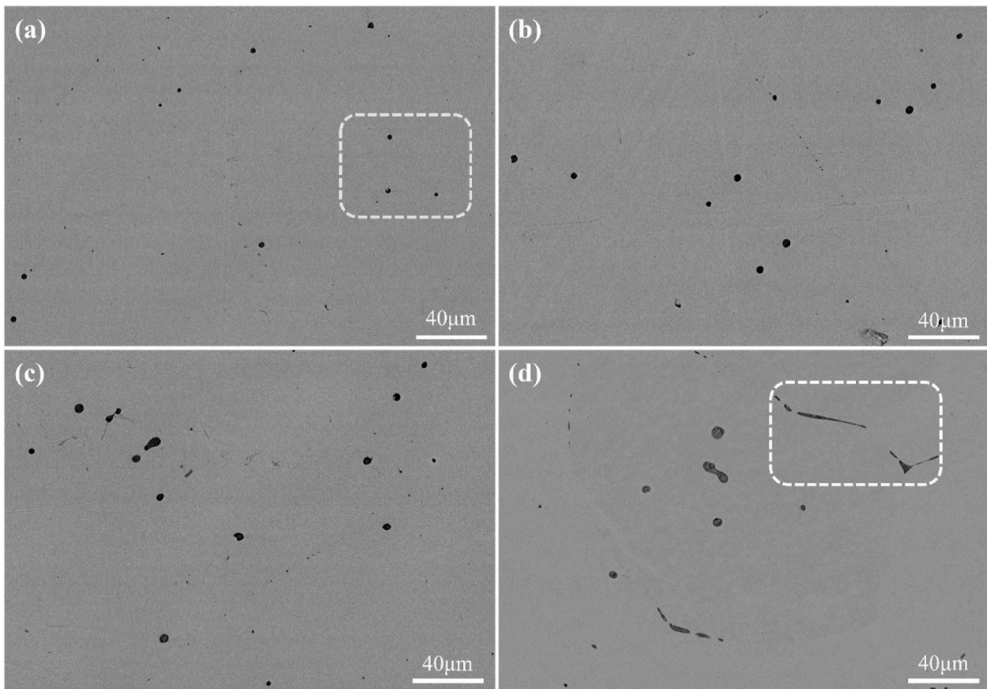


Fig. 5. SEM of Nb521–Y with various Y additions: (a) Nb521-0.1Y, (b) Nb521-0.2Y, (c) Nb521-0.3Y, (d) Nb521-0.4Y.

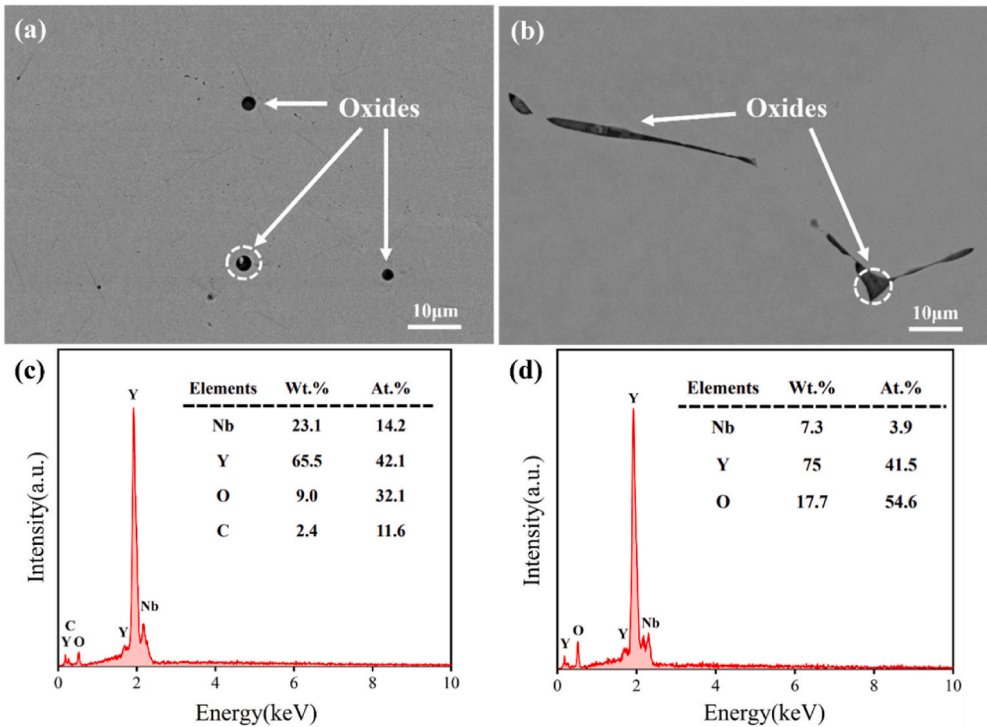


Fig. 6. High-magnification SEM and EDS results of Nb521-0.1Y and Nb521-0.4Y: (a) SEM of Nb521-0.1Y, (b) SEM of Nb521-0.4Y, (c) EDS of spherical second phase in Nb521-0.1Y, (d) EDS of acicular second phase in Nb521-0.4Y.

4. Discussion

As mentioned in 3.3 section, 0.1 wt% Y addition facilitated the YS and UTS increased by 37 and 60 MPa in Nb521-0.1Y compared to Nb521. Thus, it's necessary to explore the strengthening mechanism of Nb521-0.1Y.

Except for the yttrium oxides, the existence form of yttrium element

in Nb alloy should include solid solution. Fig. 8(a) shows the XRD of Nb521 and Nb521–Y with various Y additions. There were four peaks of Nb matrix, i.e., (110), (200), (211) and (220) peaks but no new diffraction peaks can be detected. Fig. 8(b) presented the enlarged image around (110) peak and the peaks of Nb521–Y shifted to the left compared to Nb521, which can be caused by the addition of Y. It is well known that the atomic radius of Nb (1.43 Å) is smaller than Y (1.81 Å).

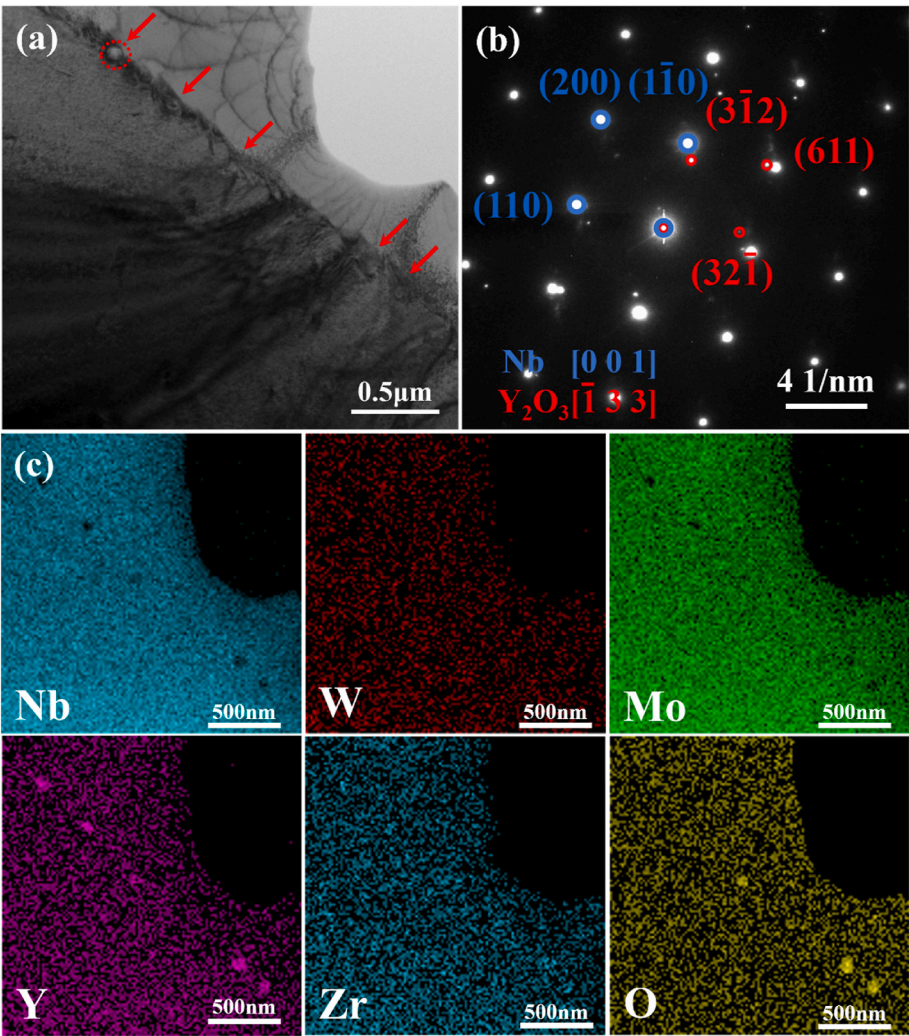


Fig. 7. TEM results of Nb521-0.1Y: (a) Bright-field TEM image, (b) SAED of the marked particle, (c) EDS mappings of Nb, W, Mo, Zr, Y and O elements.

Table 3
Measured and theoretical interplanar spacings, intersection angle of Y₂O₃.

d(Å), α (°)	d ₁ {3 1 2}	d ₂ {611}	d ₃ {32 1}	α ₁₂	α ₁₃	α ₂₃
Measured	2.85	1.73	2.80	35.0	69.9	34.9
Theoretical	2.84	1.72	2.84	34.54	69.08	34.54

Table 4
Tensile properties of Nb521 and Nb521–Y at room temperature.

	Nb521	Nb521-0.1Y	Nb521-0.2Y	Nb521-0.3Y	Nb521-0.4Y
YS, MPa	636 ± 2	673 ± 5	642 ± 6	623 ± 7	627 ± 7
UTS, MPa	680 ± 6	740 ± 7	698 ± 5	695 ± 9	679 ± 12
Elongation, %	10.8 ± 0.6	12.8 ± 1.2	13.2 ± 0.8	8.0 ± 1.5	5.9 ± 1.1

Besides, all the Nb521–Y samples exhibiting close 2θ of (100) peaks indicated that solid-soluted contents of Y were almost the same.

The contribution of solute Y atoms to YS (σ_{ss}) can be described by the following formula [19]:

$$\sigma_{ss} = \frac{A}{b} \left(\frac{F_m^4 \varepsilon}{l} \right)^{1/3} \left(\frac{c}{l^2} \right)^{2/3} \quad (1)$$

where A is a constant with a value of approximately unity, A = 1. b is magnitude of Burgers vector of Nb, b = 0.2858 nm. F_m is the maximum solute atom-dislocation interaction force. ε is the range of interaction, ε = 2d, and d is the lattice spacing of Nb, thus ε = 0.66 nm. l is dislocation line tension and is approximately μb²/2. c is solute concentration, c = 0.001. l is the mean spacing of solute Y.

F_m can be calculated using the following formula [20]:

$$F_m = \frac{\mu b^2}{\phi} \varepsilon_L \quad (2)$$

where μ is shear modulus of Nb, μ = 37.5 GPa. φ is a numerical constant, φ = 6. ε_L is the misfit parameter, and is defined as following formula [19]:

$$\varepsilon_L = \left[\left(\frac{\varepsilon_\mu}{1 + 0.5|\varepsilon_\mu|} \right)^2 + (\alpha \varepsilon_b)^2 \right]^{2/3} \quad (3)$$

where ε_μ is the shear modulus misfit parameter (∂μ/∂c × 1/μ), and for the system of Y atoms in Nb, ε_μ = -0.32. ε_b is the lattice misfit parameter (∂b/∂c × 1/b), and for the system of Y atoms in Nb, ε_b = 0.28. α is a constant, α = 15.

The mean spacing of Y in Nb can be estimated using the following formula [21]:

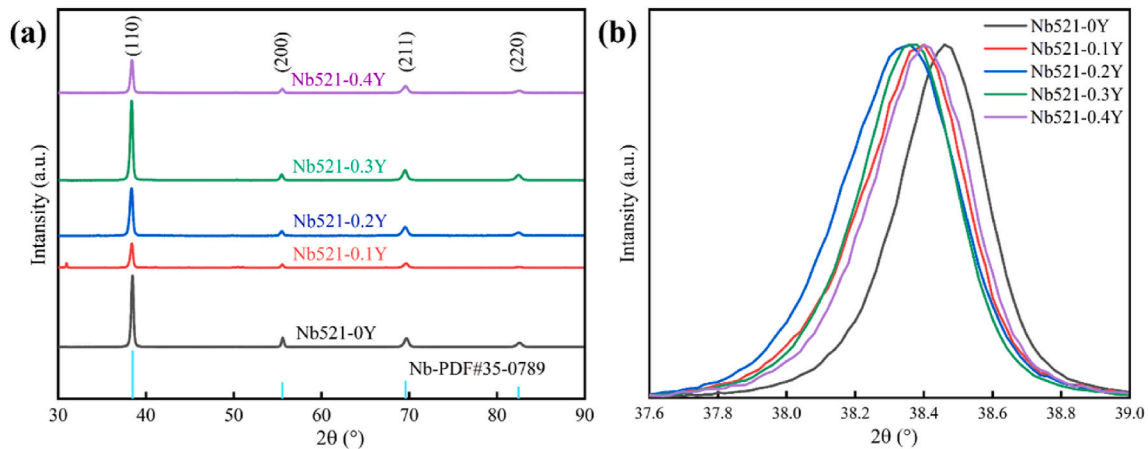


Fig. 8. (a) XRD of Nb521 and Nb521–Y with various Y additions, (b) Enlarged image around (110) peak.

$$l = \left(\frac{2}{c^{1/3}} - 3 \right) b_{\text{mtx}} + 2 \left(\frac{b_{\text{mtx}} + b_{\text{sol}}}{2} \right) \quad (4)$$

where b_{mtx} is magnitude of Burgers vector for matrix material Nb, $b_{\text{mtx}} = 0.2858$ nm. b_{sol} is magnitude of Burgers vector for solute Y, $b_{\text{sol}} = 0.3647$ nm.

Basing on the formulas (1–4), the calculated contribution of solute Y atoms to YS in Nb521–0.1Y is about 15 MPa.

The contribution to YS by nano Y_2O_3 (σ_p) can be defined using the following Orowan strengthening formula [22]:

$$\sigma_p = M \cdot 0.045 \frac{\mu b}{\lambda} \ln \frac{r}{b} \quad (5)$$

where M is the Taylor factor, and $M = 3.06$ in BCC metal. λ is the mean inter-particle spacing, $\lambda = 360$ nm. r is particle radius, $r = 40$ nm. Thus, the calculated contribution of Y_2O_3 to YS in Nb521–0.1Y is about 20 MPa.

Briefly, the calculated total contribution of solute Y atoms and Y_2O_3 to YS in Nb521–0.1Y was about 35 MPa, which was close to the measured YS difference of 37 MPa between Nb521 and Nb521–0.1Y. In future, more and smaller yttrium oxides will be introduced in Nb521 to achieve higher strength.

5. Conclusion

To solve the low strength of niobium (Nb) alloy, yttrium (Y) element was introduced into Nb matrix to achieve solution strengthening from Y atoms and dispersion strengthening from yttrium oxides. The latter one was realized by oxygen carrier route and vacuum electromagnetic levitation melting technique. Five kinds of Nb alloys including Nb521, Nb521–0.1Y, Nb521–0.2Y, Nb521–0.3Y, Nb521–0.4Y were prepared for comparison. Then grain size, second phases, especially yttrium oxides were characterized by LSCM, FE-SEM and TEM. Finally, tensile properties of these Nb alloys were tested and the solution strengthening of Y atoms and Orowan strengthening of yttrium oxides were evaluated quantitatively. The main conclusion drawn from the results can be summarized as follows:

- (1) Y addition facilitated the grain refinement of Nb521. Furthermore, Nb521–0.1Y and Nb521–0.2Y displayed smaller grain size. The grain size was 423, 152, 103, 187 and 224 μm for Nb521, Nb521–0.1Y, Nb521–0.2Y, Nb521–0.3Y and Nb521–0.4Y, respectively.
- (2) The Y-bearing second phases in Nb521–Y was Y_2O_3 , and the size of which increased with Y additions. In Nb521–0.1Y, the size of

Y_2O_3 exhibited bimodal distribution characteristics, i.e., nano-scale of 80 nm and micron-scale of 1.6 μm .

- (3) Compared to Nb521, the Nb521–0.1Y exhibited higher YS of 673 MPa, UTS of 740 MPa and larger elongation of 12.8%. The YS, UTS, and elongation of Nb521 were 636 MPa, 680 MPa and 10.8%, respectively.
- (4) The total calculated contribution of solute Y atoms and nano-scale Y_2O_3 to yield strength in Nb521–0.1Y was about 35 MPa, which was very close to the measured difference of 37 MPa.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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